

Lecture 2, Dec 18th, 2025.

Properties: i). $|a \cdot b| = |a| \cdot |b|$

ii). $\left| \frac{a}{b} \right| = \frac{|a|}{|b|}$ if $b \neq 0$.

iii). $|a^n| = |a|^n$

Now assume $a > 0$ then.

i). $|x| = a$ if and only if $x = \pm a$. ✓

ii). $|x| < a$ if and only if $-a < x < a$. ($-a < x$ and $x < a$) ✓

iii). $|x| > a$ if and only if $x > a$ or $x < -a$

Problem. Solve $x^3 + 3x^2 > 4x$.

Sln: Step 1. Unequal to zero.

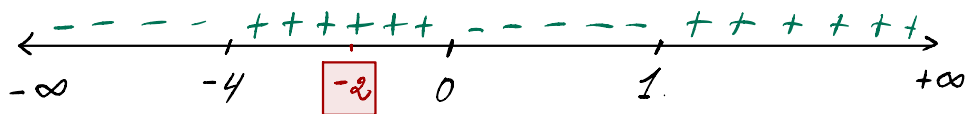
$$x^3 + 3x^2 - 4x > 0.$$

Step 2. Factorize.

$$x(x^2 + 3x - 4) > 0$$

$$x(x + 4)(x - 1) > 0$$

Step 3. Draw a line with the roots of the polynomial above.



Step 4. Select "easy" number and evaluate the sign

if $x = -2$ $(-2)(-2+4)(-2-1) = (-2)(2)(-3) = +12$.

Step 5. Select the region for your problem.

$$x(x+4)(x-1) > 0 \quad (\text{positive numbers})$$

$$\mathcal{S} = (-4, 0) \cup (1, +\infty).$$

Example. iii). Solve $|3x+2| \geq 4$ $\stackrel{\text{def}}{\Rightarrow} 3x+2 \geq 4$ or $3x+2 \leq -4$

$$3x \geq 4-2 \quad 3x \leq -4-2$$

$$3x \geq 2 \quad 3x \leq -6$$

$$x \geq \frac{2}{3} \quad x \leq -\frac{6}{3}$$

then $S_1 = [\frac{2}{3}, \infty)$ $S_2 = (-\infty, -2]$

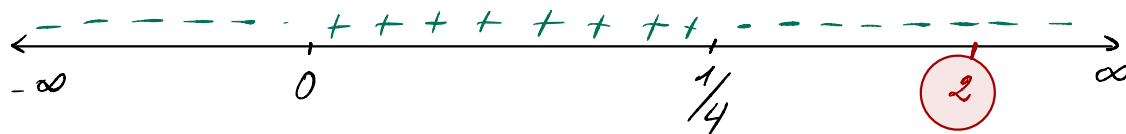
$$S = S_1 \cup S_2 = (-\infty, -2] \cup [\frac{2}{3}, \infty)$$

Problem. Solve $\frac{1}{x} < 4$.

Sln: $\frac{1}{x} - \frac{4}{1} < 0$.

$$\frac{1-4x}{x} < 0$$

$$\frac{1-4(2)}{2} = \frac{1-8}{2} = -\frac{7}{2}$$



$$S = (-\infty, 0) \cup (\frac{1}{4}, \infty)$$

Problems. i). Prove the triangular inequality.

For all $a, b \in \mathbb{R}$

$$|a+b| \leq |a| + |b|$$

ii). Solve for x , assuming that a, b , and c are negative constants.

Sln: i). Let $a, b \in \mathbb{R}$ and notice that.

$$-|a| \leq a \leq |a|$$

$$-|b| \leq b \leq |b|$$

$$-|a| - |b| \leq a+b \leq |a| + |b|$$

$$-(\underbrace{|a| + |b|}_{-s}) \leq \underbrace{a+b}_x \leq \underbrace{|a| + |b|}_s$$

by definition $|a+b| \leq |a| + |b|$

ii). $ax + b < c$ Solving for x .

$$ax < c - b.$$

$$x > \frac{c-b}{a} \quad \text{b/c } a \text{ is negative number.}$$

$$\mathcal{S} = \left(\frac{c-b}{a}, \infty \right).$$

Limit of a function (Intuitive).

What is the result of $\frac{0}{0}$?

$$\frac{1}{1} = 1 \quad \frac{2}{2} = 1 \quad \frac{10}{10} = 1$$

If you can try define $\frac{0}{0} = 1$ then

$$1 = \frac{0}{0} = \frac{2 \times \cancel{0}}{\cancel{0}} = 2.$$

Since $1 \neq 2$ then, there is not correct definition of $\frac{0}{0}$, that is, $\frac{0}{0}$ is not a number.

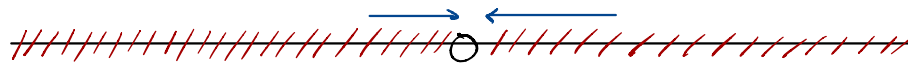
Sometimes the functions present this problem.

$$\text{Let } f(x) = \frac{x-1}{x^2-1}.$$

$$\text{Notice that if } x \neq 1 \text{ then } f(x) = \frac{(x-1)}{(x-1)(x+1)} = \frac{1}{x+1}$$

if $x=1$, what is $f(1)$?

Notice that is not possible compute $f(1)$, but we can ask yourself about what is "approximately" $f(1)$.



For values of x near to 1 we have.

$x < 1$	$f(x)$	$x > 1$	$f(x)$
0.5	0.666667	1.5	0.400000
0.9	0.526316	1.1	0.476190
0.99	0.502513	1.01	0.497512
0.999	0.500250	1.001	0.499750
0.9999	0.500025	1.0001	0.499975



1



0.5



1



0.5

We see that the closer x is to 1 (on either sides of 1) the closer $f(x)$ is 0.5. In fact, it appears that we can make values of $f(x)$ as close as we like to 0.5 by taking x sufficiently close to 1.

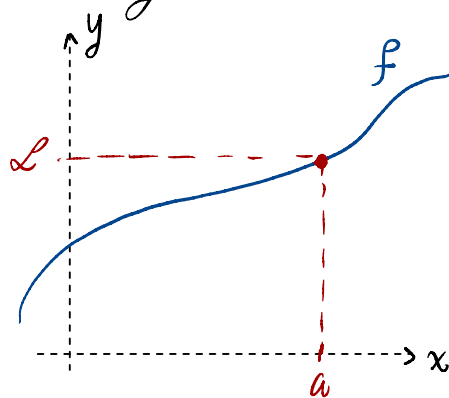
We express this by saying "the limit of the function $f(x) = \frac{x-1}{x^2-1}$ as x approaches 1 is equal to 0.5" or.

$$\lim_{x \rightarrow 1} \frac{x-1}{x^2-1} = 0.5$$

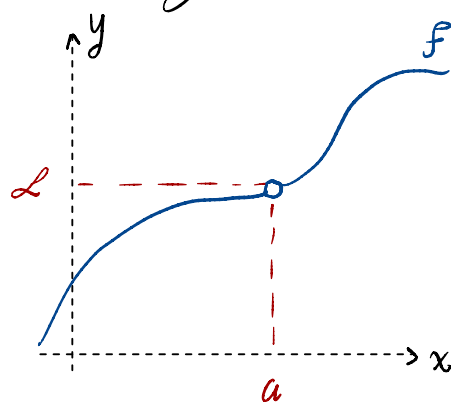
Def. (Intuitive Definition of Limit) Suppose that the function f is defined on some open interval that contains $\underline{a}$, x (except possibly at $\underline{a}$ itself) then write

$$\lim_{x \rightarrow a} f(x) = L$$

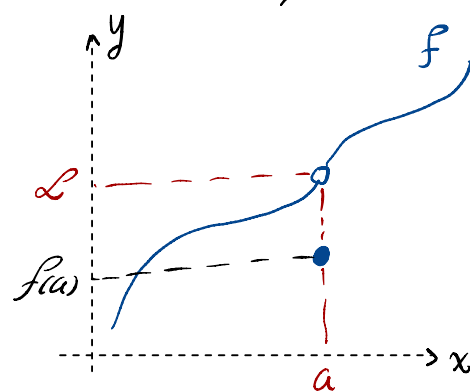
If we can make the values of $f(x)$ arbitrarily close to L by restricting x to be sufficiently close to a . But not equal to $\underline{a}$.



(I)



(II)



(III)

In (I) $\lim_{x \rightarrow a} f(x) = L$ and $f(a) = L$.

In (II) $\lim_{x \rightarrow a} f(x) = L$ but $f(a)$ is not defined.

In (III) $\lim_{x \rightarrow a} f(x) = L$, $f(a)$ is defined but $f(a) \neq L$

Example. Estimate the value of $\lim_{t \rightarrow 0} \frac{\sqrt{t^2+9} - 3}{t^2}$.

Soln: if $t = \pm 1$ $f(\pm 1) = \frac{\sqrt{1+9} - 3}{1} = \sqrt{10} - 3 \approx 0.1622$.

$$\text{if } t = \pm \frac{1}{2} \quad f\left(\pm \frac{1}{2}\right) = \frac{\sqrt{\frac{1}{4} + 9} - 3}{\frac{1}{4}} = \frac{\sqrt{\frac{37}{4}} - 3}{\frac{1}{4}} \approx 0.1655.$$

$\vdots$

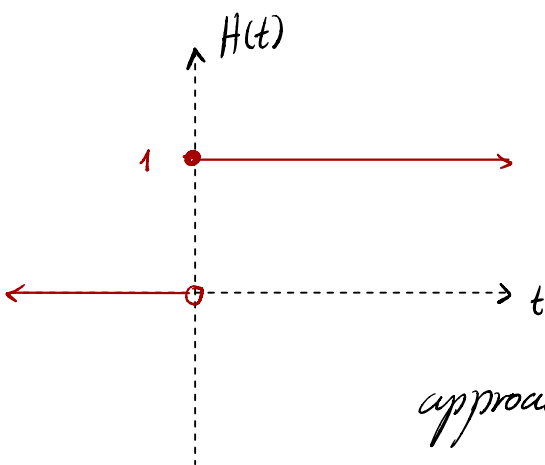
$$\text{if } t = \pm \frac{1}{10} \quad \text{then } f\left(\pm \frac{1}{10}\right) = \frac{\sqrt{\frac{1}{100} + 9} - 3}{\frac{1}{100}} \approx 0.1666..$$

So, when $t \rightarrow 0$ then $f(x) \rightarrow 0.1666... = \frac{1}{6}$.

In the next class, we can check this using the rules.

Example. (for you) Guess the value of $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

One-sided Limits Let consider.



$$H(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0. \end{cases}$$

$\lim_{t \rightarrow 0} H(t)$ does not exist because

there is not a single value that $H(t)$ approaches as t approaches to 0.

However, as t approaches to 0 from the left $H(t)$ approaches to 0.
and as t approaches to 0 from the right $H(t)$ approaches to 1.

So, we can write this

$$\lim_{t \rightarrow 0^-} H(t) = 0$$

left side

and

$$\lim_{t \rightarrow 0^+} H(t) = 1$$

right side.

Def. (Intuitive Definition of one-sided limits) We write

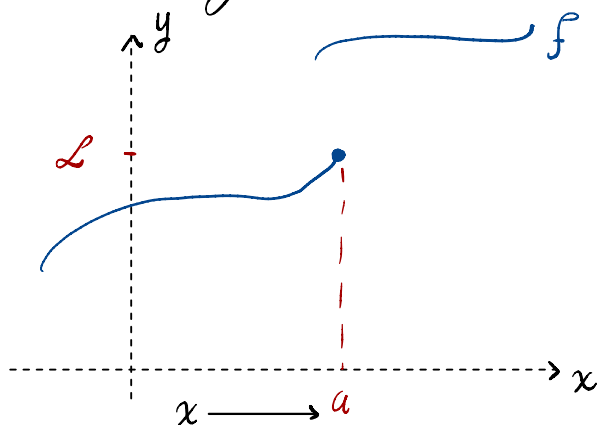
$$\lim_{x \rightarrow a^-} f(x) = L.$$

and say that the left-hand limit of $f(x)$ as $x \rightarrow a$ is equal to L if we can make the values of $f(x)$ arbitrary close to L by restricting x to be sufficiently close to a with $x < a$.

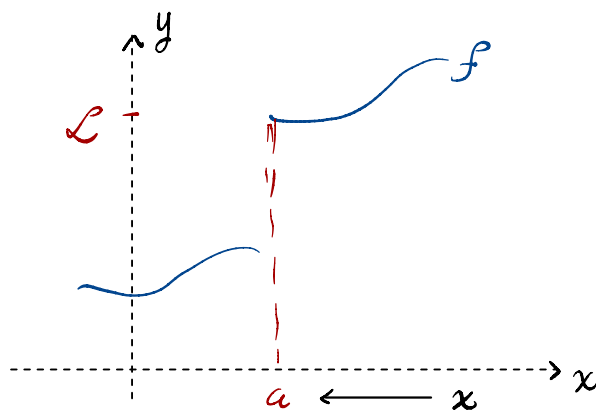
Similarly, we write

$$\lim_{x \rightarrow a^+} f(x) = L$$

and call it right-hand limit.



$$\lim_{x \rightarrow a^-} f(x) = L.$$



$$\lim_{x \rightarrow a^+} f(x) = L.$$

Property ♥

$\lim_{x \rightarrow a} f(x) = L$ if and only if $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L.$

Example:

compute:

a). $\lim_{x \rightarrow 2^-} g(x) = 3.$

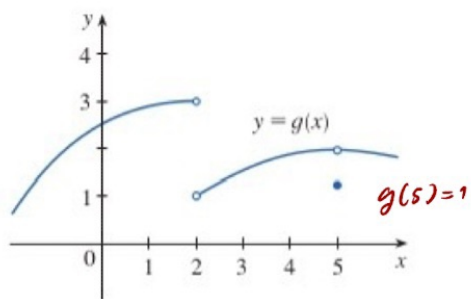
b). $\lim_{x \rightarrow 2^+} g(x) = 1.$

c). $\lim_{x \rightarrow 2} g(x) = \text{DNE}$

d). $\lim_{x \rightarrow 5^-} g(x) = 2$

e). $\lim_{x \rightarrow 5^+} g(x) = 2$

f). $\lim_{x \rightarrow 5} g(x) = 2$



Example. Find $\lim_{x \rightarrow 0} \frac{1}{x^2}$ if it exists.

Sln. As x becomes close to 0, x^2 becomes close to 0, and $\frac{1}{x^2}$ becomes very large.

For example

$$\text{if } x = \pm \frac{1}{2} \text{ then } f\left(\pm \frac{1}{2}\right) = \frac{1}{\frac{1}{4}} = 4.$$

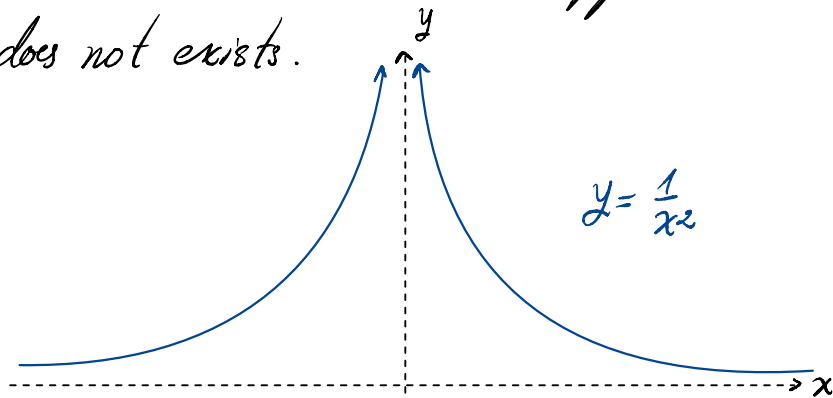
$$\text{if } x = \pm \frac{1}{5} \text{ then } f\left(\pm \frac{1}{5}\right) = \frac{1}{\frac{1}{25}} = 25$$

$$\text{if } x = \pm \frac{1}{10} \text{ then } f\left(\pm \frac{1}{10}\right) = \frac{1}{\frac{1}{100}} = 100$$

$$\text{if } x = \pm 0.001 \text{ then } f\left(\pm \frac{1}{1000}\right) = 1,000,000.$$

In fact, the values of $f(x)$ can be made arbitrary large by taking x close to 0. Thus $f(x)$ do not approach a number, so

$\lim_{x \rightarrow 0} \frac{1}{x^2}$ does not exist.

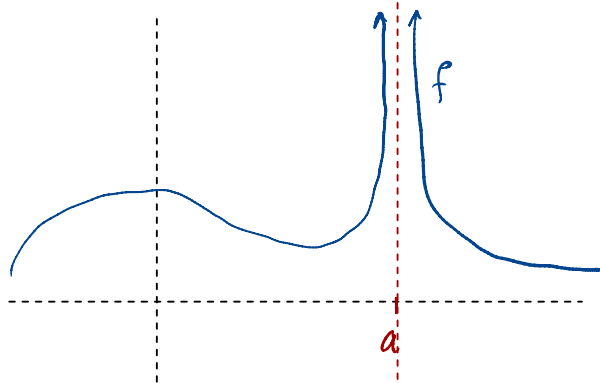


Infinite Limits; Vertical Asymptotes.

To indicate the kind of behavior exhibited in above example we use the notation $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$.

Def. (Intuitive Definition of an infinite limit) Let f be a function defined on both sides of a , except possibly at a itself. Then $\lim_{x \rightarrow a} f(x) = \infty$ means that the values of $f(x)$ can be made large

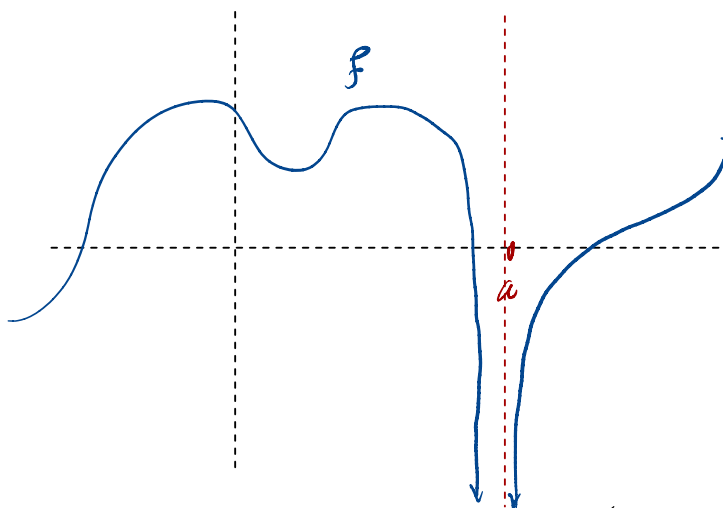
by taking x values close to a , but not equal to a .



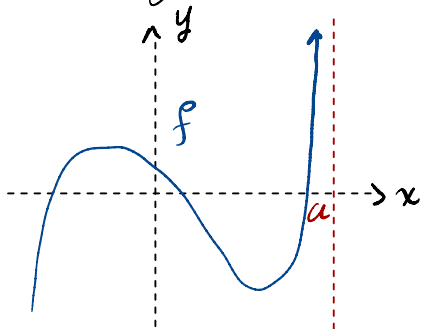
$$\lim_{x \rightarrow a} f(x) = \infty.$$

Similarly, for function as

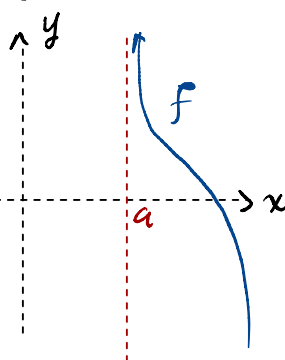
$$\lim_{x \rightarrow a} f(x) = -\infty.$$



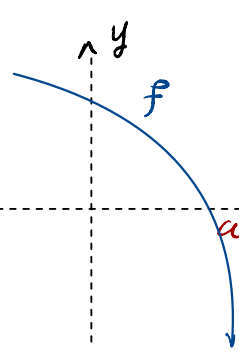
Similarly definitions can be given for one-sided infinite limits.



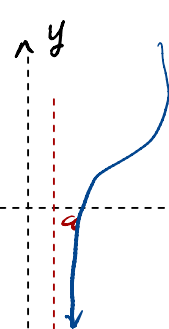
$$\lim_{x \rightarrow a^-} f(x) = \infty$$



$$\lim_{x \rightarrow a^+} f(x) = \infty$$



$$\lim_{x \rightarrow a^-} f(x) = -\infty$$



$$\lim_{x \rightarrow a^+} f(x) = -\infty$$

Def. The vertical line $x=a$ is called a vertical asymptote of the curve $y=f(x)$ if at least one of the following statements is true.

$$\lim_{x \rightarrow a} f(x) = \infty$$

$$\lim_{x \rightarrow a^-} f(x) = \infty$$

$$\lim_{x \rightarrow a^+} f(x) = \infty$$

$$\lim_{x \rightarrow a} f(x) = -\infty$$

$$\lim_{x \rightarrow a^-} f(x) = -\infty$$

$$\lim_{x \rightarrow a^+} f(x) = -\infty.$$

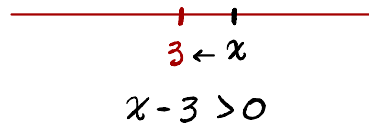
Example. Does the curve $y = \frac{2x}{x-3}$ have a vertical asymptote?

In other words is there any number a s.t

$$\lim_{x \rightarrow a} f(x) = \pm \infty.$$

Sln. There is a potential vertical asymptote where the denominator is 0, that is at $x=3$. So, we investigate the one-sided limits here.

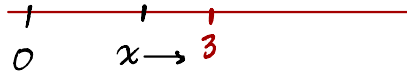
$$\lim_{x \rightarrow 3^+} \frac{2x}{x-3} = +\infty.$$



if $x > 3$ then $x-3 > 0$ and

$$2x > 0 \text{ then } \frac{2x}{x-3} > 0$$

$$\lim_{x \rightarrow 3^-} \frac{2x}{x-3} = -\infty.$$



if $x < 3$ then $x-3 < 0$

and $x > 0$ b/c x is close to 3 then $2x > 0$ so

$$\frac{2x}{x-3} < 0$$